MUSTAFA AVCU

SOFTWARE ENGINEER | MACHINE LEARNING | GENERATIVE AI | AWS

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PROFESSIONAL SUMMARY

Software engineer with 4 years of experience at Amazon building scalable backend services, event-driven data processing pipelines and reusable machine learning infrastructure across advertising and customer service chatbot domains. Hands-on AWS experience with strong Python, Java, Scala, SQL skills and a deep learning background. Built guardrails for Amazon's first LLM-powered customer service chatbot and the ML infrastructure behind a customer issue prediction model that cut average chatbot interaction time by 9%.

TECHNICAL SKILLS

Languages: Python, Java, Scala, TypeScript, SQL, JavaScript, C++
Cloud & Infrastructure: AWS, CDK, CloudFormation, Lambda, Step Functions, S3, DynamoDB, SQS, SNS
Data Engineering: Spark, Glue, EMR, ETL data pipelines, AWS Clean Rooms
AI/ML: Large Language Models (LLMs), Generative AI, guardrails, Keras, TensorFlow, SageMaker, Bedrock
Engineering: CI/CD, Docker, Git, infrastructure as code, microservices, REST APIs, on-call support

PROFESSIONAL EXPERIENCE

Amazon | Software Development Engineer II

Seattle, WA & Toronto, ON | August 2022 – June 2026

- Implemented a guardrail layer in a new LLM orchestration service connecting generative AI models to Amazon's customer service chatbot; validated customer inputs and model outputs to ensure safe, appropriate interactions, enabling the launch of Amazon's first LLM-powered customer support experience.
- Designed and built reusable ML infrastructure using Amazon SageMaker and CI/CD pipelines to automate model training, testing, and deployment for customer service chatbot workflows.
- Deployed a machine learning model using the infrastructure above to infer customer issues and route them to relevant support workflows, reducing average chatbot interaction time by 9% (440 seconds to 400 seconds) with no negative impact on other metrics.
- Owned Retailer Ad Service's attribution integration with Amazon Demand-Side Platform (ADSP), expanding beyond Sponsored Products (SP) to support display and video ads; built unified ETL data pipelines and developed attribution logic in Spark SQL within AWS Clean Rooms to jointly perform attribution across both ad programs.
- Redesigned the Retailer Ad Service's catalog ingestion architecture to support catalogs with 2M+ offers refreshed every two hours; root-caused 300ms+ upsert latency to redundant updates from misordered category normalization; fixed the update check to skip unchanged offers without breaking downstream obsolete offer detection workflow.
- Reduced OpenSearch ingestion traffic for the catalog service by 90% by filtering out timestamp-only DynamoDB change events, winning the production impact award at an Amazon organization-wide hackathon.

Agency for Science, Technology and Research (A*STAR)

Singapore | Data Science Intern | June 2018 – September 2018

- Developed SeizNet, a convolutional neural network for seizure detection using EEG data from 29 patients; published at IEEE ICASSP 2019 and presented in Brighton, UK.
- Achieved 95.8% sensitivity at 0.17 false alarms per hour; a 2-channel configuration achieved 93.3% sensitivity and outperformed a traditional SVM baseline using all EEG channels.

EDUCATION

University of Southern California — M.S. in Computer Science | CGPA: 3.96/4.00 | May 2022
Selected coursework: Machine Learning, Analysis of Algorithms, Database Systems, Applied Machine Learning for Games, Estimation Theory, Random Processes, Applied Natural Language Processing

Bilkent University — B.S. in Electrical and Electronics Engineering | CGPA: 3.93/4.00 | June 2019

HONORS & AWARDS

- Two-time Amazon organization-wide hackathon winner among 20+ competing teams
- Annenberg Fellowship with admission to the USC PhD program (2019)
- IEEExtreme 10.0 (2016): 2nd in Turkey, 20th in IEEE Region 8, 65th global
- IEEExtreme 9.0 (2015): 3rd in Turkey, 30th in IEEE Region 8, 82nd global
- Nationwide University Entrance Exam (LYS): ranked in the top 300 out of 2 million students in Turkey (2014)

PUBLICATION

M. T. Avcu, Z. Zhang, and DWS. Chan, "Seizure Detection Using Least EEG Channels by Deep Convolutional Neural Networks" IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2019.